

Assignment 1

Harthik Py

Sol^m | To prove that all col^m of $\sum_{i=1}^m x_i = 0$ is subspace of $\mathbb{R}^n$

Step-1 Φ

Assume let $U = \{x \in \mathbb{R}^n : \sum_{i=1}^m x_i = 0\}$, $V = \mathbb{R}^n$

To prove : U is subspace of $\mathbb{R}^n$

Proof : Since $x \in \mathbb{R}^n$ & U is set of n vectors drawn from $\mathbb{R}^n$ that pass a condition

$\therefore U \subseteq \mathbb{R}^n$

Step 2 To prove : $U \neq \emptyset$

Assume same st definition from Step 1

Proof : $\sum_{i=1}^n x_i \Rightarrow x \sum_{i=1}^n 0 = 0+0+\dots+0 = 0$ (null vector)

$\therefore 0$ Satisfy the condition definition of $U \Rightarrow 0 \in U$
 $U \neq \emptyset$ [Non-empty vector set]

Step 3 Closure under Addition [inner operⁿ]

To prove : $x, y \in U \Rightarrow x+y \in U$
if x, y belong to U then their sum belong U

Assume: Let $x, y \in U$, arbitrary

Now : By definition of U

$$x \in U$$

$$y \in U$$

∴ h_n satisfies the definiteness condⁿ of U .

Thus $h_n \in U$ i.e. U is closed under multiplication.

Hence U is a subspace of R^n .

Sol^m 2 Given $Ax=b$, A is 3×5 , $b = [60, 60, 36]^T$

Augmented Matrix

$$\left[\begin{array}{ccccc|c} 1 & 2 & 3 & 4 & 6 & 60 \\ 1 & 1 & 7 & 4 & 4 & 60 \\ 1 & 5 & 4 & 2 & 1 & 36 \end{array} \right]$$

$$R_2 \rightarrow R_2 - R_1, R_3 \rightarrow R_3 - R_1$$

$$\left[\begin{array}{ccccc|c} 1 & 2 & 3 & 4 & 6 & 60 \\ 0 & -1 & 4 & 0 & -2 & 0 \\ 0 & 3 & 1 & -2 & -5 & -24 \end{array} \right]$$

$$R_3 \rightarrow R_3 + 3R_2, R_2 \rightarrow R_2 \times (-1)$$

$$\left[\begin{array}{ccccc|c} 1 & 2 & 3 & 4 & 6 & 60 \\ 0 & 1 & -4 & 0 & 2 & 0 \\ 0 & 0 & 13 & -2 & -11 & -24 \end{array} \right] \rightarrow \text{REF}$$

Now convert to RREF (Backward pass)

$$R_3 \rightarrow R_3 / 13,$$

$$\left[\begin{array}{ccccc|c} 1 & 2 & 3 & 4 & 6 & 60 \\ 0 & 1 & -4 & 0 & 2 & 0 \\ 0 & 0 & 1 & -2/13 & -11/13 & -24/13 \end{array} \right]$$

$$R_1 \rightarrow R_1 - 3R_3, R_2 \rightarrow R_2 + 4R_3$$

$$\left[\begin{array}{ccccc|c} 1 & 2 & 0 & 58/13 & 11/13 & 60 + 72/13 \\ 0 & 1 & 0 & 8/13 & -18/13 & -96/13 \\ 0 & 0 & 1 & -2/13 & -11/13 & -24/13 \end{array} \right]$$

Row 1

$$R_1$$

$$0, (1-2)(A-3)$$

$$(4-4), (4-6)$$

$$R_3$$

$$0, 5-2, (A-3)$$

$$(2-4), (-6)$$

$$R_3$$

$$0, 0, 1$$

$$(1+3 \times 4)$$

$$-2, -5-6$$

$$4 + \frac{6}{13} \Rightarrow 52 + \frac{24}{13}$$

$$-11 \frac{3}{13} - 3 \times \frac{6}{13}$$

$$-13 \frac{3}{13}$$

$$28 + 33$$

$$\frac{121}{13}$$

$$6 + 3 \times \frac{11}{13}$$

$$28 \frac{33}{13}$$

$$\frac{111}{13}$$

$$2 - 4 \times \frac{11}{13}$$

$$6 - 44 \frac{3}{13} = \frac{28}{13}$$

$$2 + 4 \times \frac{11}{13}$$

$$6 - 44 \frac{3}{13}$$

$$\frac{28}{13}$$

$$R_1 \sim R_2$$

$$\begin{array}{cccccc|c} 1 & 0 & 0 & 74/13 & 147/13 & 1 & 1044/13 \\ 0 & 1 & 0 & -8/13 & -18/13 & 1 & -96/13 \\ 0 & 0 & 1 & -2/13 & -11/13 & 1 & -24/13 \end{array} \rightarrow \text{RREF}$$

$$u_3 = -\frac{24}{13} + \frac{11}{13}$$

$$= \frac{11}{13}$$

Now take $u_4 = u_5 = 0$ Particular solⁿ — (i)

$$u_3 = -24/13$$

$$u_2 = -96/13$$

$$u_1 = 1044/13$$

$$u_p = (1/13) [1044, -96, -24, 0, 0]^T$$

Homogeneous System $Ax = 0$ — (ii)

$$u_1 + 74/13 u_4 + 147/13 u_5 = 0$$

$$u_2 - \frac{8}{13} u_4 - \frac{18}{13} u_5 = 0$$

$$u_3 - \frac{2}{13} u_4 - \frac{11}{13} u_5 = 0$$

$$\Rightarrow u_1 = -\frac{1}{13} (74u_4 + 147u_5)$$

$$\Rightarrow u_2 = \frac{1}{13} (8u_4 + 18u_5)$$

$$\Rightarrow u_3 = \frac{1}{13} (2u_4 + 11u_5)$$

$$u_h = \begin{bmatrix} (-74u_4 - 147u_5)/13 \\ (8u_4 + 18u_5)/13 \\ (2u_4 + 11u_5)/13 \\ u_4 \\ u_5 \end{bmatrix}$$

$$\Rightarrow u_4 \begin{bmatrix} -74/13 \\ 8/13 \\ 2/13 \\ 1 \\ 0 \end{bmatrix} + u_5 \begin{bmatrix} -147/13 \\ 18/13 \\ 11/13 \\ 0 \\ 1 \end{bmatrix}$$

u_4, u_5 are arbitrary real numbers.

$\Rightarrow h_1, h_2$ can be substituted absorbing $1/13$

$$\therefore \Rightarrow u_h = h_1 [-74, 8, 2, 13, 0]^T + h_2 [-147, 18, 11, 0, 13]^T$$

$$h_1, h_2 \in \mathbb{R}$$

General Solⁿ — (11)

$$\text{From } A \cdot u_p = b \text{ \& From } A \cdot u_h = 0$$

$$A(u_p + u_h) = A u_p + A u_h = b + 0 = b$$

$$\therefore u_g = (1/13) [104, -96, -24, 0, 0]^T + h_1 [-74, 8, 2, 13, 0]^T + h_2 [-147, 18, 11, 0, 13]^T, h_1, h_2 \in \mathbb{R}$$

Sol^m 3

$$C = \begin{bmatrix} 1 & 2 & 1 & 2 & -1 \\ 1 & 1 & 0 & -1 & 4 \\ 1 & -2 & 4 & 1 & 0 \end{bmatrix} \quad \begin{array}{l} C \text{ is } 3 \times 5 \\ R^{3,5} \end{array}$$

(A) For Column of C

Step-1 $C_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ $C_2 = \begin{bmatrix} 2 \\ 1 \\ -2 \end{bmatrix}$ $C_3 = \begin{bmatrix} 1 \\ 0 \\ 4 \end{bmatrix}$ $C_4 = \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}$ $C_5 = \begin{bmatrix} -1 \\ 4 \\ 0 \end{bmatrix}$

To prove: $U = \text{col}(C)$ is subspace of $\mathbb{R}^3$

So, proof: each C_i has 3 entries so $C_i \in \mathbb{R}^3$

These are 5 scalar quantity in each.

take $(a_1, a_2, a_3, a_4, a_5) \in \mathbb{R}$, arbitrary.

so, it form = $U = \left\{ u \in \mathbb{R}^3 : u = \sum_{i=1}^5 a_i C_i, \right\} a \in \mathbb{R}$

$$= \text{span}(C_1, C_2, C_3, C_4, C_5) \quad \forall \in \mathbb{R}^3$$

$\therefore \underline{U} \subset \mathbb{R}^3$

Step 2

To prove $U \neq \emptyset$

Assume: definition of U in step 1

proof: Take $\vec{a}_1, \vec{a}_2, \vec{a}_3, \vec{a}_4, \vec{a}_5 = \vec{0}$

$$0c_1 + 0c_2 + \dots + 0c_5 = \vec{0}$$

Thus: $\vec{0} \in U$.

$\therefore U \neq \emptyset$ [Non empty vector set]

Step 3

(A) closure under Addition [inner opⁿ]

To prove: $u, v \in U \Rightarrow u+v \in U$

$$u, v \in U \Rightarrow u+v \in U$$

Assume: $u, v \in U$, arbitrary -

proof:

$$u \in U \Rightarrow u = a_1c_1 + a_2c_2 + \dots + a_5c_5 = \sum_{i=1}^5 a_i c_i$$

$$v \in U \Rightarrow v = b_1c_1 + b_2c_2 + \dots + b_5c_5 = \sum_{i=1}^5 b_i c_i$$

$$a_i, b_i \in R$$

$$\begin{aligned} u+v &= \sum_{i=1}^5 (u+v)_i \Rightarrow \sum_{i=1}^5 u_i + \sum_{i=1}^5 v_i \\ &\Rightarrow \sum_{i=1}^5 a_i c_i + \sum_{i=1}^5 b_i c_i \end{aligned}$$

$$\Rightarrow \sum_{i=1}^5 (a_i + b_i) c_i$$

let $d_i = a_i + b_i i$ since $a_i, b_i \in \mathbb{R}$ & $\mathbb{R}$ is closed under Addition $d_i \in \mathbb{R}$

So, $u + v = \sum d_i c_i$ with $d_i \in \mathbb{R}$

Thus $u + v \in U$ i.e. U is closed under addition

(A)

Scalar

Step 4.4 Closure under multiplication [outer opⁿ]

To prove: $u \in U, \lambda \in \mathbb{R} \Rightarrow \lambda u \in U$.

Assume: let $u \in U$, arbitrary & $\lambda \in \mathbb{R}$

proof:

$$\Rightarrow u \in U \Rightarrow \sum_{i=1}^s a_i c_i \rightarrow \text{From step 3} \quad a_i \in \mathbb{R}$$

$$\Rightarrow \lambda \cdot u \Rightarrow \sum_{i=1}^s \lambda (a_i c_i) \Rightarrow \sum_{i=1}^s (\lambda a_i) c_i$$

$\Rightarrow$ Let $e_i \in U$ $e_i = \lambda a_i$. Since $\lambda \in \mathbb{R}$ and $\mathbb{R}$ is closed under multiplication.
 $e_i \in \mathbb{R}$

$$\text{So, } \lambda u = \sum_{i=1}^s e_i c_i \text{ with } e_i \in \mathbb{R}$$

Thus $\lambda u \in U$, i.e. U is closed under scalar multiⁿ

Hence $U \neq \emptyset$, U is closed under Addition
& U is closed under Scalar multiⁿ
 $\therefore \text{cd}(U)$ is subspace of $\mathbb{R}^3$

(A)
Step 2:

To prove: which column of C are linearly independent.

proof: $C = \begin{bmatrix} 1 & 2 & 1 & 2 & -1 \\ 1 & 1 & 0 & -1 & 4 \\ 1 & -2 & 4 & 1 & 0 \end{bmatrix} \begin{matrix} R_1 \\ R_2 \\ R_3 \end{matrix}$

$$R_2 \leftarrow R_2 - R_1, R_3 \leftarrow R_3 - R_1$$

$$\begin{bmatrix} 1 & 2 & 1 & 2 & -1 \\ 0 & (-1-2) & (0-1) & (-1-2) & 4-(-1) \\ 0 & (-2-2) & (4-1) & (1-2) & 0-(-1) \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & 2 & 1 & 2 & -1 \\ 0 & -1 & -1 & -3 & 5 \\ 0 & -4 & 3 & -1 & 1 \end{bmatrix}$$

$$R_3 \leftarrow R_3 - 4R_2$$

$$\begin{bmatrix} 1 & 2 & 1 & 2 & -1 \\ 0 & -1 & -1 & -3 & 5 \\ 0 & (-4+4) & 3-(-4) & -1-(-3) & 1-20 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & 2 & 1 & 2 & -1 \\ 0 & -1 & -1 & -3 & 5 \\ 0 & 0 & 7 & 11 & -19 \end{bmatrix}$$

REF

Pivot are 1, -1, 7 in col 1, 2, 3

$\therefore C_1, C_2, C_3$ are linearly independent.

$\text{Span}(C_1, C_2, C_3)$

~~Step 6~~ Take c_1, c_2, c_3 only remove non pivot
Finally ~~Rec.~~

$$\text{Basis of } U = \begin{bmatrix} 1 & 2 & 1 \\ 1 & 1 & 0 \\ 1 & -2 & 4 \end{bmatrix}$$

$$\dim(U) = 3.$$

∴ For part A: dimension of $U = 3$

$$\therefore U = \mathbb{R}^3$$

~~Ans.~~

(B) Space spanned by Row of C

$$\text{Step 1: } r_1 = (1, 2, 1, 2, -1)$$

$$r_2 = (1, 1, 0, -1, 4)$$

$$r_3 = (1, -2, 4, 1, 0)$$

To prove $W = \text{Row}(C)$ is a ^{set} subspace of $\mathbb{R}^5$

$$\text{So let } W = \{ w \in \mathbb{R}^5 : w = a_1 r_1 + a_2 r_2 + a_3 r_3, a_i \in \mathbb{R} \}$$

$$\begin{aligned} V &= \mathbb{R}^5 \\ &= \text{Span}(r_1, r_2, r_3) \end{aligned}$$

Thus $\forall w \in \mathbb{R}^5$ so every

thus : each $r_i \in R^5$, & R^5 is closed under addition & scalar multiplication

So, every combination $(a_1 r_1 + a_2 r_2 + a_3 r_3)$ will fall in R^5

$$\therefore W \subseteq R^5.$$

(B)

Step 1.2

To prove: $W \neq \emptyset$

Proof take $a_1 = a_2 = a_3 = \dots = 0$

$$\sum_{i=1}^3 a_i r_i = a_1 r_1 + a_2 r_2 + a_3 r_3 = 0$$

Thus $0 \in W \therefore W \neq \emptyset$

(B)

Step 1.3

To prove $w_1, w_2 \in W \Rightarrow w_1 + w_2 \in W$

Assume $w_1, w_2 \in W$, arbitrary

Proof :

$$w_1 \in W \Rightarrow w_1 = \sum_{i=1}^3 a_i r_i, a_i \in R$$
$$w_2 \in W \Rightarrow w_2 = \sum_{i=1}^3 b_i r_i, b_i \in R$$

$$w_1 + w_2 = \sum_{i=1}^3 a_i r_i + \sum_{i=1}^3 b_i r_i$$

$$= \sum_{i=1}^3 (a_i + b_i) r_i$$

Let $d_i = a_i + b_i, d_i \in R$

$$\Rightarrow w_1 + w_2 = \sum_{i=1}^3 d_i r_i \rightarrow \text{satisfy the condition}$$

Thus $w_1 + w_2 \in W$, i.e. W is closed under addition.

(B)

Step 4

To prove $w \in W, \lambda \in R \Rightarrow \lambda w \in W$

Assume: $\lambda \in R$ arbitrary & $w \in W$ arbitrary

proof $\lambda \cdot w = \sum \lambda (a_i r_i), a_i \in R$

$$= \sum_{i=1}^3 (\lambda a_i) r_i$$

$$= (\lambda a_1 + \lambda a_2 + \lambda a_3) r_3$$

$$= \lambda a_1 r_1 + \lambda a_2 r_2 + \lambda a_3 r_3$$

Let $e_i = \lambda a_i \in R$. So $e_i \in R$

$$\lambda w = \sum e_i r_i$$

Thus $\lambda w \in W$, i.e. W is closed under scalar multiplication

Hence $W = \text{Row}(C)$ is a subspace of $\mathbb{R}^5$

(B)

Step 2 REF form A Step 2

$$\begin{bmatrix} 1 & 2 & 1 & 2 & -1 \\ 0 & -1 & -1 & -3 & 5 \\ 0 & 0 & 7 & 11 & -19 \end{bmatrix}$$

Thus $\text{Rank}(3)$ & x_1, x_2, x_3 are LI

Finally:

$$\text{Basis of } W = \begin{bmatrix} 1 & 2 & 1 & 2 & -1 \\ 0 & -1 & -1 & -3 & 5 \\ 0 & 0 & 7 & 11 & -19 \end{bmatrix}$$

$$\dim(W) = 3$$

~~A~~

$$A + B \Rightarrow \dim U = \dim W = 3 = \text{rank}(C) \\ (\text{row rank} = \text{column rank})$$

$$\text{So, } v_1 = \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix} \quad v_2 = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix} \quad w_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \quad w_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$$

Q4 Similar pattern :

Part (a) Subspace proof

Step 1 - subset $\Rightarrow S_1 \cap S_2 \subseteq \mathbb{R}^3$ (To prove)

Step 2 - Non-emptiness $\Rightarrow 0 \in S_1 \cap S_2$

Step 3 - Closed under $+$ $\Rightarrow x+y \in S_1 \cap S_2$

Step 4 - Closure under $\cdot$ $\Rightarrow \lambda x \in S_1 \cap S_2$

Hence Step 1 $\rightarrow$ Step 4 Subspace

part (b) basis & dimensions

Step 5 - Membership equation $\rightarrow Mz = 0$

Step 6 - Row-Reduce M , Rank-nullity

Step 7 - Recover the vector, verify both ways

Step 8 - Basis & Dim. find that.

(A) To prove $S_1 \cap S_2$ is subspace of $\mathbb{R}^3$

Step 1: To prove $S_1 \cap S_2 \subseteq \mathbb{R}^3$ (subset)

Given $S_1 = \text{span}\{v_1, v_2\}$ & $S_2 = \text{span}\{w_1, w_2\}$

Proof so ~~$u \in \mathbb{R}^3$~~ every $v_i, w_i \in \mathbb{R}^3$ & span of vector of $\mathbb{R}^3$ is a subspace of $\mathbb{R}^3$ thus S_1 & S_2 are subspaces of $\mathbb{R}^3$

$$S_1 \cap S_2 = \{ u \in \mathbb{R}^3, u \in S_1 \text{ and } u \in S_2 \} \quad V = \mathbb{R}^3$$

$\Rightarrow u \in S_1 \cap S_2 \Rightarrow u \in S_1$ & $u \in S_2$

$\Rightarrow S_1 \subseteq \mathbb{R}^3$ & $S_2 \subseteq \mathbb{R}^3$

Step 2: To prove $0 \in S_1 \cap S_2$ that is $S_1 \cap S_2 \neq \emptyset$

Let S_1 and S_2 are subspaces

proof:

$$\begin{aligned} S_1, S_2 \text{ are subspaces} &\Rightarrow 0 \in S_1 \text{ and } 0 \in S_2 \\ &\Rightarrow 0 \in S_1 \cup S_2 \end{aligned}$$

$$\therefore S_1 \cap S_2 \neq \emptyset$$

Step 3: To prove $u, y \in S_1 \cap S_2 \Rightarrow u+y \in S_1 \cap S_2$

let $u, y \in S_1 \cap S_2$, [u, y arbitrary]

proof:

$$\begin{aligned} u \in S_1 \cap S_2 &\Rightarrow u \in S_1 \text{ and } u \in S_2 \\ y \in S_1 \cap S_2 &\Rightarrow y \in S_1 \text{ and } y \in S_2 \end{aligned}$$

$$\begin{aligned} u \in S_1 &\Rightarrow u = a_1 v_1 + a_2 v_2, \quad a_1, a_2 \in \mathbb{R} \\ y \in S_2 &\Rightarrow y = b_1 v_1 + b_2 v_2, \quad b_1, b_2 \in \mathbb{R} \end{aligned}$$

$$\begin{aligned} u+y &= a_1 v_1 + a_2 v_2 + b_1 v_1 + b_2 v_2 \\ &= (a_1+b_1)v_1 + (a_2+b_2)v_2 \end{aligned}$$

$$\text{let } d_1 = a_1+b_1, \quad d_2 = a_2+b_2 \text{ since } a_1, a_2, b_1, b_2 \in \mathbb{R}$$

$$d_1, d_2 \in \mathbb{R}$$

$$\begin{aligned} \Rightarrow u+y &= d_1 v_1 + d_2 v_2, \quad d_1, d_2 \in \mathbb{R} \\ \Rightarrow u+y &\in S_1 \cap S_2 \end{aligned}$$

Step 4 To prove: $u \in S_1 \cap S_2, \lambda \in \mathbb{R} \Rightarrow \lambda u \in S_1 \cap S_2$

Assumption: $u \in S_1$ & $u \in S_2, \lambda \in \mathbb{R}$

$$\text{proof: } \lambda u = \lambda(a_1 v_1 + a_2 v_2) = (\lambda a_1) v_1 + (\lambda a_2) v_2$$

$$\Rightarrow \lambda u \in S_1$$

$$\lambda u = \lambda(a_3 w_1 + a_4 w_2) = (\lambda a_3) w_1 + (\lambda a_4) w_2$$

$$= \lambda u \in S_2$$

thus $\therefore \lambda u \in S_1 \cap S_2$

Hence $\therefore S_1 \cap S_2$ is subspace of $\mathbb{R}^3$

(B) Basis & dim.

$$u \in S_1 \cap S_2 \Rightarrow a_1 v_1 + b_2 v_2 \text{ and } u = c w_1 + d w_2$$

Since $u = u$ [For two vector set]

$$\Rightarrow a_1 v_1 + b_2 v_2 - c w_1 - d w_2 = 0$$

$$\Rightarrow a_1 \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix} + b_2 \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix} - c \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} - d \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} = 0$$

$$M = \begin{bmatrix} 1 & 1 & -1 & 0 \\ 0 & 2 & -1 & -1 \\ 2 & 2 & 0 & -1 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - R_1$$

$$M = \begin{bmatrix} 1 & 1 & -1 & 0 \\ 0 & 2 & -1 & -1 \\ 0 & 0 & 0 & -1 \end{bmatrix} \text{ - REF}$$

$$\text{Rank}(M) = 3$$

Basic: a, b, c Free: d

$$\text{Rank} + \text{Nullity} = \text{No. of Rows Col.}$$

$$\text{Rank} - \text{Nullity} = 4 - 3 \quad \therefore \text{Rank} + \text{Nullity} = \text{No. of Col.}$$

$$\text{Rank} = 3$$

$$\text{Col} = 4$$

$$\text{Null} = 1$$

Back substitution

$$R_3: 2c - d = 0 \Rightarrow d = 2c \Rightarrow c = d/2$$

$$R_2: 2b - c - d = 0 \Rightarrow b = 3d/4$$

$$R_1: a + b - c = 0 \Rightarrow a = c - b = d/4 - 3d/4 = -d/2$$

$$\text{Take } d = 4 \Rightarrow a = -1, b = 3, c = 2, d = 4$$

Verify Both ways:

$$\text{via } s_1: -1(1, 0, 2)^T + 3(1, 2, 2)^T = (2, 6, 4)^T$$

$$\text{via } s_2: 2(1, 1, 0)^T + 4(0, 1, 1)^T = (2, 6, 4)^T$$

$$\text{Divide by 2: } u = (1, 3, 2)^T$$

$$\text{Basis of } s_1 \cap s_2 = \{ (1, 3, 2)^T \}$$

$$\dim(s_1 \cap s_2) = 1 = \text{nullity}$$

Solⁿ 5

$$v_1 = (1, 1, 0, 0)^T \quad w_1 = (0, 0, 1, 1)^T$$

$$v_2 = (0, 1, 1, 0)^T \quad w_2 = (0, 0, 0, 1)^T$$

$$S_1 = \text{Span}\{v_1, v_2\} \quad S_2 = \text{Span}\{w_1, w_2\}$$

$$S_1 + S_2 = \{x + y : x \in S_1, y \in S_2\}$$

(a) Prove $S_1 + S_2$ is a subspace of $\mathbb{R}^4$

Step 1 Assume $S_1 + S_2 = \{x \in \mathbb{R}^4 : x = y + z\}$
 $y \in S_1, z \in S_2$

$$V = \mathbb{R}^4$$

proof:

$$y \in S_1 \subseteq \mathbb{R}^4$$

$$z \in S_2 \subseteq \mathbb{R}^4$$

$$\mathbb{R}^4 \text{ closed under } + \Rightarrow y + z \in \mathbb{R}^4$$

$$\text{Thus } S_1 + S_2 \subseteq \mathbb{R}^4$$

Step 2: To prove: $S_1 + S_2 \neq \emptyset$

$$\text{proof: } S_1 \text{ subspace} \Rightarrow 0 \in S_1$$

$$S_2 \text{ subspace} \Rightarrow 0 \in S_2$$

$$0 + 0 \Rightarrow 0 \text{ with } 0 \in S_1, S_2$$

$$\text{Thus: } 0 \in S_1 + S_2 \therefore S_1 + S_2 \neq \emptyset$$

Step-3 To prove: $u_1 + u_2 \Rightarrow$

$$u_1, u_2 \in S_1 + S_2 \Rightarrow u_1 + u_2 \in S_1 + S_2$$

(inner opⁿ) (+)

Assume $u_1, u_2 \in S_1 + S_2$ arbitrary

~~to prove~~

$$u_1 = y_1 + z_1, y_1 = a_1 v_1 + a_2 v_2 \in S_1$$

$$z_1 = a_3 w_1 + a_4 w_2 \in S_2$$

$$u_2 = y_2 + z_2, y_2 = b_1 v_1 + b_2 v_2 \in S_1$$

$$z_2 = b_3 w_1 + b_4 w_2 \in S_2$$

proof:

$$u_1 + u_2 = (y_1 + z_1) + (y_2 + z_2)$$

$$= (y_1 + y_2) + (z_1 + z_2)$$

$$\Rightarrow y_1 + y_2 = (a_1 + b_1) v_1 + (a_2 + b_2) v_2$$

$$= d_1 v_1 + d_2 v_2 \in S_1$$

$$\Rightarrow z_1 + z_2 = (a_3 + b_3) w_1 + (a_4 + b_4) w_2$$

$$= d_3 w_1 + d_4 w_2 \in S_2$$

where $d_i = a_i t b_i \in R$ (R closed under $+$)

$$u_1 + u_2 = \underbrace{d_1 v_1 + d_2 v_2}_{\in S_1} + \underbrace{d_3 w_1 + d_4 w_2}_{\in S_2}$$

Thus: $u_1 + u_2 \in S_1 + S_2$

Step 4 To prove $u_1 \in S_1 + S_2, h \in R \Rightarrow h u_1 \in S_1 + S_2$
(outer opⁿ)($\circ$)

Assume: $u_1 = y_1 + z_1, y_1 = a_1 v_1 + a_2 v_2 \in S_1$
 $z_1 = a_3 w_1 + a_4 w_2 \in S_2$

proof:

$$h u_1 = h(y_1 + z_1) = h y_1 + h z_1$$

$$h y_1 = h(a_1 v_1 + a_2 v_2) = (h a_1) v_1 + (h a_2) v_2 \\ = e_1 v_1 + e_2 v_2 \in S_1$$

$$h z_1 = h(a_3 w_1 + a_4 w_2) = (h a_3) w_1 + (h a_4) w_2 \\ = e_3 w_1 + e_4 w_2 \in S_2$$

where $e_i = h a_i \in R$ (closed under $\times$)

$$h u_1 = \underbrace{(e_1 v_1 + e_2 v_2)}_{\in S_1} + \underbrace{(e_3 w_1 + e_4 w_2)}_{\in S_2}$$

$\therefore h u_1 \in S_1 + S_2$

Hence $S_1 + S_2 \neq \emptyset$, Closed under $+$, $\lambda \cdot$

Axioms inherited from $\mathbb{R}^4$

∴ $S_1 + S_2$ is subspace of $\mathbb{R}^4$

b) Basis & dimension

Step 5: $S_1 + S_2 = \text{Span}\{v_1, v_2, w_1, w_2\}$

$$M = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix}$$

REF

Step 6 $R_2 \leftarrow R_2 - R_1, R_3 \rightarrow R_3 - R_2; R_4 \rightarrow R_4 - R_3$

$$M = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

4 pivot in col 1, 2, 3, 4 $\Rightarrow$ rank = 4

All four independent.

Step 7

Basis = pivot column of original M

$$= \{(1, 1, 0, 0)^T, (0, 1, 0, 0)^T, (0, 0, 1, 1)^T, (0, 0, 0, 1)^T\}$$

$$\dim(S_1 + S_2) = 4$$